

By: AKH
 Rev.--- By: ---
 Chk: ---

Date Jun 2026
 Date ---
 Date ---

MacTest

Macaulay Beam Analysis - Example Worksheet

104 - (2) Span - Point UDL

Beam Sect = W14x61 W14x90
 A = 17.90 26.50 in²
 I = 640.00 999.00 in⁴

E = 29,000,000 lb / in²

Length Units in
 Force Units lb

Beam Segments (First Segment Begins at 0)

	Segment No.	X End	A	I	E
		in	in ²	in ⁴	lb / in ²
DO NOT DELETE ROWS (GRAPHED)	1	900.00	17.90	640.00	2.900E+07
	2				
	3				
	4				
	5				

Supports

	Supp. No.	X	Type	Imposed Movement	
		in	---	* dY (vert)	** dr (rot)
DO NOT DELETE ROWS (GRAPHED)	1	0.0	P	0.0	0.000
	2	300.0	P	0.0	0.000
	3	900.0	P	0.0	0.000
	4				
	5				

* (settlement) ** (Not Working)

Distributed Loads (downward loads entered as NEGATIVE) - UNFACTORED

	Load No.	x1	x2	w1 (at x1)	w2 (at x2)
		in	in	lb / in	lb / in
DO NOT DELETE ROWS (GRAPHED)	1	0.0	900.0	-5.1	-5.1
	2	0.0	0.0	0.0	0.0
	3	0.0	0.0	0.0	0.0
	4	0.0	0.0	0.0	0.0
	5	0.0	0.0	0.0	0.0

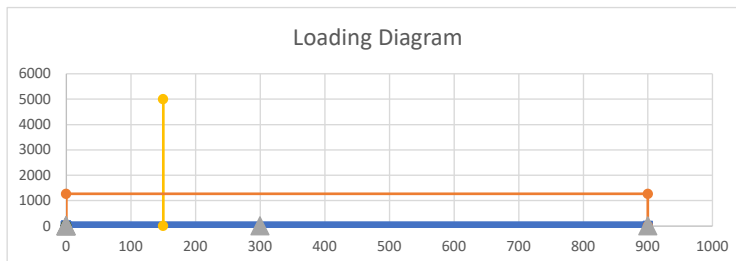
Load
lb / ft
61.0
0.0
0.0
0.0

Point Loads - UNFACTORED

	Load No.	X	Py	My
		in	lb	lb-in
DO NOT DELETE ROWS (GRAPHED)	1	150	-5000.00	0.00
	2	0	0.00	0.00
	3			
	4			
	5			

My Load
lb-ft
0.0
0.0
0.0

Loading Diagram



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BEAM REACTIONS

Position (X)	Enforced Disp (dy)	Enforced Rot (theta)	Vert Reac (Ry)	Internal Moment (Mz)	Type
(L)	(L)	(rad)	(F)	(F-L)	(---)
0.00	0.0	0.0	2,378.2	0.0	P
300.00	0.00	0.00	6,114.0	0.0	P
900.00	0.00	0.00	1,082.8	0.0	P

Vert Reac
(kip)
2.378
6.114

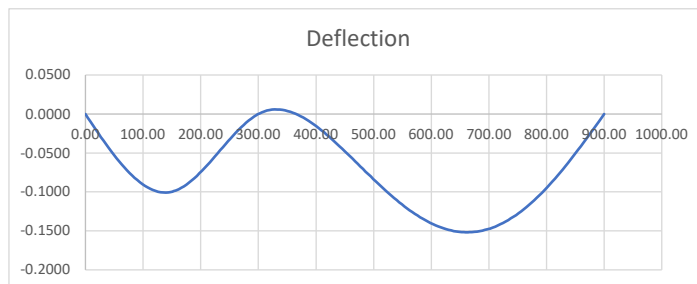
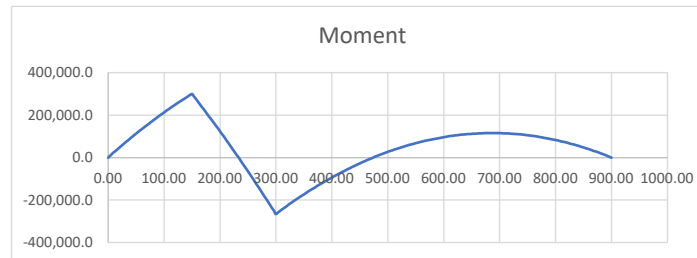
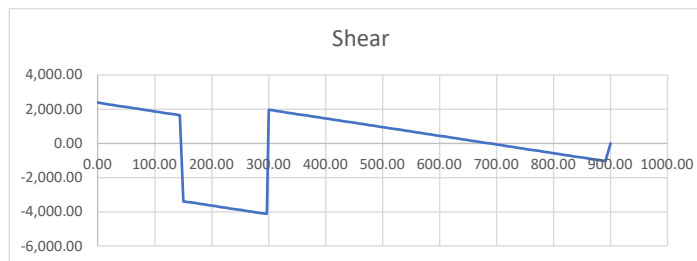
BEAM REACTIONS EQUILIBRIUM CHECK

Item	Units	Loads	Reactions	Residual	Check Sum
Sum Fy	F	-9,575.0	9,575.0	0.0	OK
Sum M @ x=0	F-L	-2.8088E+06	2.8088E+06	0.0	OK

BEAM MAX/MIN LOAD EFFECTS

Item	Position (X)	Shear (V)	Moment (M)	Defl. (delta)
(---)	(L)	(F)	(F-L)	(L)
Max +V	0.00	2,378.2	0.0	0.0000
Max -V	297.00	-4,131.6	-252,873.0	-0.0012
Max +M	150.00	-3,384.3	299,542.2	-0.0998
Max -M	300.00	1,967.2	-265,290.6	0.0000
Max Defl	666.00	106.7	114,215.2	-0.1518

Max/Min Value
(---)
2.38 kip
-4.13 kip
24.96 kip-ft
-22.11 kip-ft
-0.152 in



INPUT SUMMARY

General Info

Beam Length:	75 ft
Section Name:	W14x61
Self Weight:	False

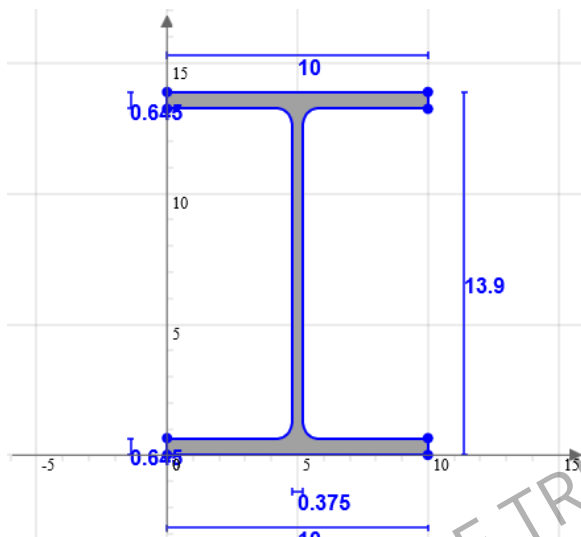
Supports

Support Type	Location
Pinned	0 ft
Roller	75 ft
Roller	25 ft

Loads

Load Type	Location	Magnitude	Load Case
Point Load	12.5 ft	-5000 lb	DL
Distributed Load	0 ft to 75 ft	-61 lb to -61 lb	DL

Beam Section



Geometric Properties		
A	17.9	in ²
C _z	5	in
C _y	6.95	in

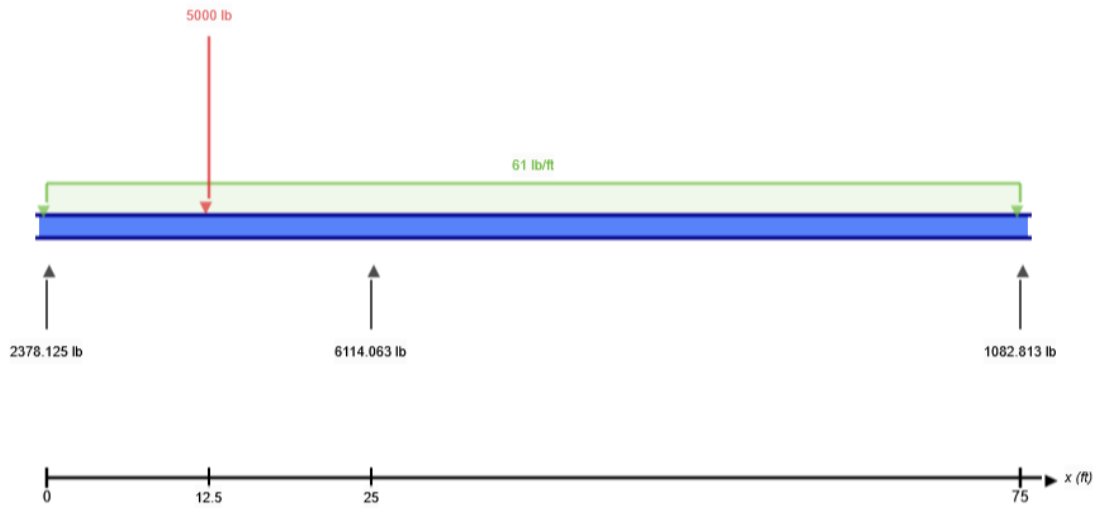
Bending Properties		
I _z	640	in ⁴
I _y	107	in ⁴

Shear Properties		
A _z	11.277	in ²
A _y	4.99	in ²

Torsion Properties		
J	2.19	in ⁴
r _r	0.991	in

Shape	Material	E (ksi)	v	ρ (lb/ft ³)
W14x61	Structural Steel	29000	0.27	490

FREE BODY DIAGRAM



RESULT SUMMARY

Check	Status	Limit	Ratio	Max
Deflection	PASS	L/250	0.063	L/3961
Custom Stress Limit	PASS	39 ksi	0.083	3.253 ksi
Material Yield	PASS	36 ksi	0.09	3.253 ksi
Material Strength	PASS	58 ksi	0.056	3.253 ksi

ANALYSIS RESULTS

Reactions

Support at	R _x (lb)	R _y (lb)	M (kip-ft)
0	0	2378.125	0
25	0	6114.063	0
75	0	1082.813	0

Force Extremes

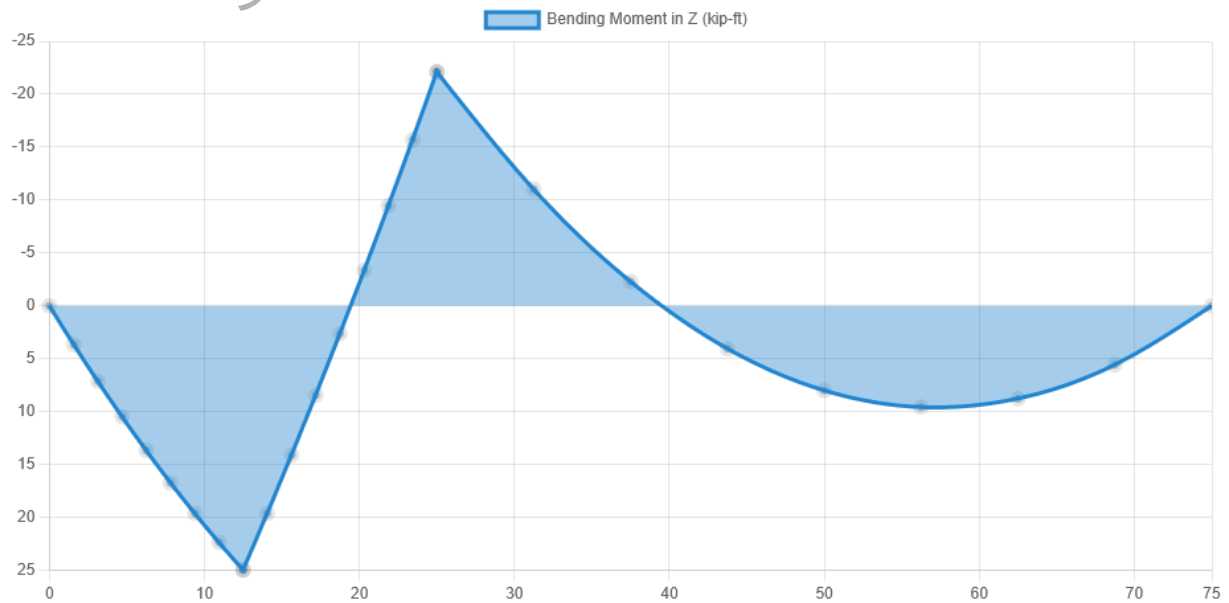
Result	Max	Min
Bending Moment	24.961 kip-ft	-22.109 kip-ft
Shear	2378.125 lb	-4146.875 lb
Displacement Y	0 in	-0.151 in

Stress Extremes

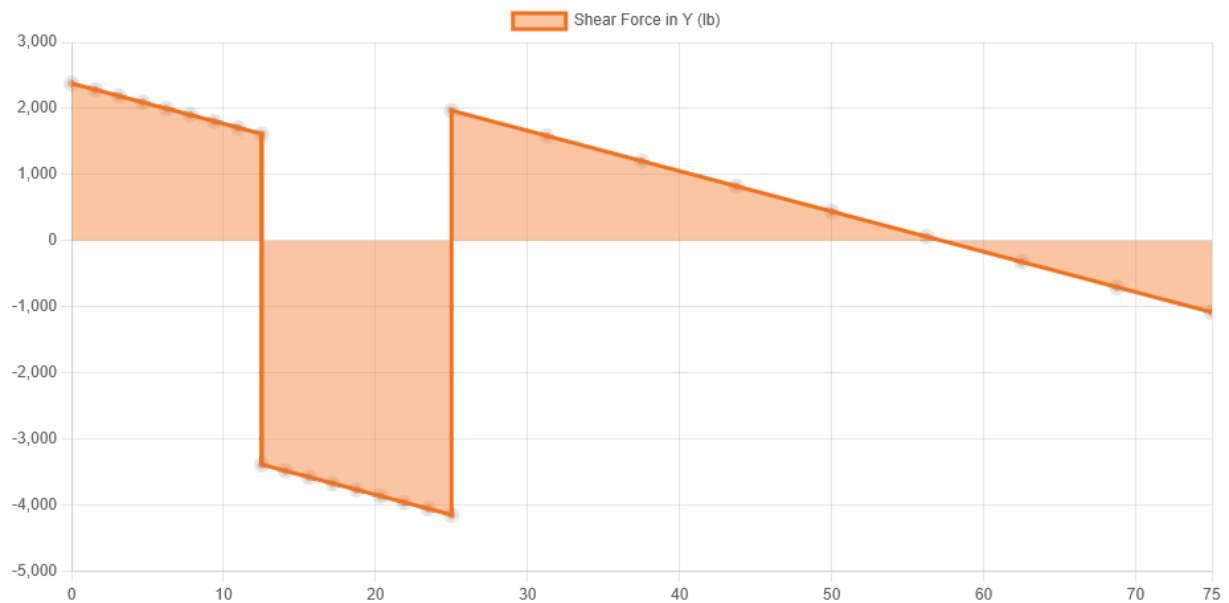
Result	Max (ksi)	Min (ksi)
Bending Stress	3.253	-3.253
Shear Stress Total	0.885	0.013
Max Combined Normal Stress	3.253	0
Min Combined Normal Stress	0	-3.253

DIAGRAMS

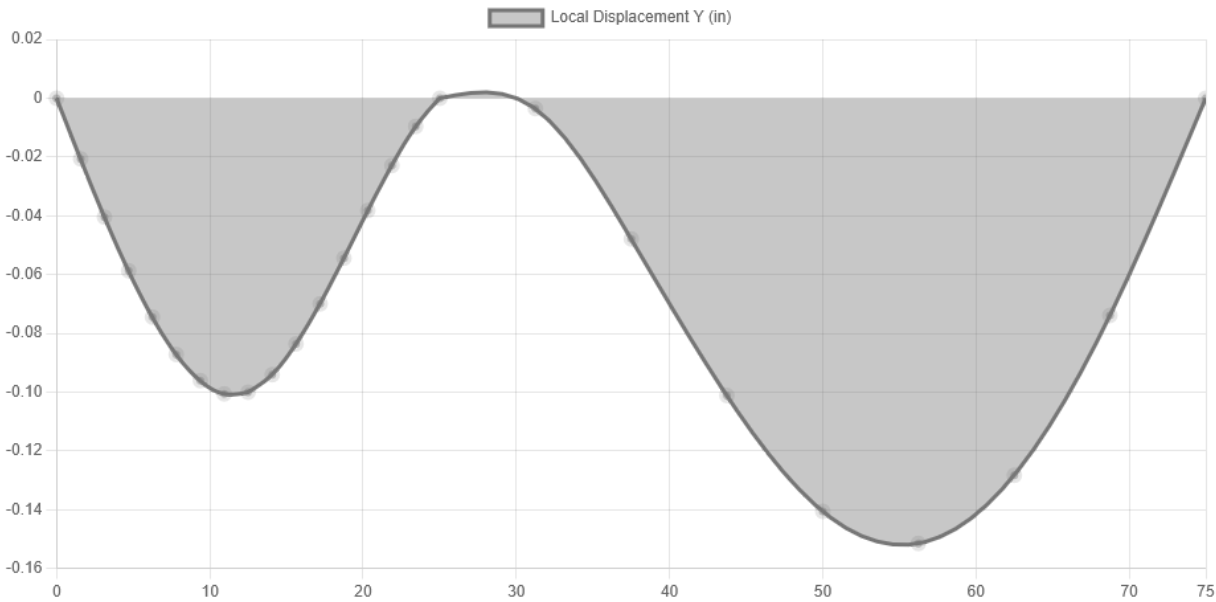
Bending Moment Diagram



Shear Force Diagram



Displacement



Location (ft)	0	10.937	25	56.25	75
Total Deflection (in)	0 ⓘ	0.101 ⓘ	0 ⓘ	0.151 ⓘ	0 ⓘ
Deflection/Span	-	L/2981	-	L/3961	-
ⓘ The Deflection/Span results are calculated using the analysis results and the Deflection Limit of L/250 set in the model settings.					